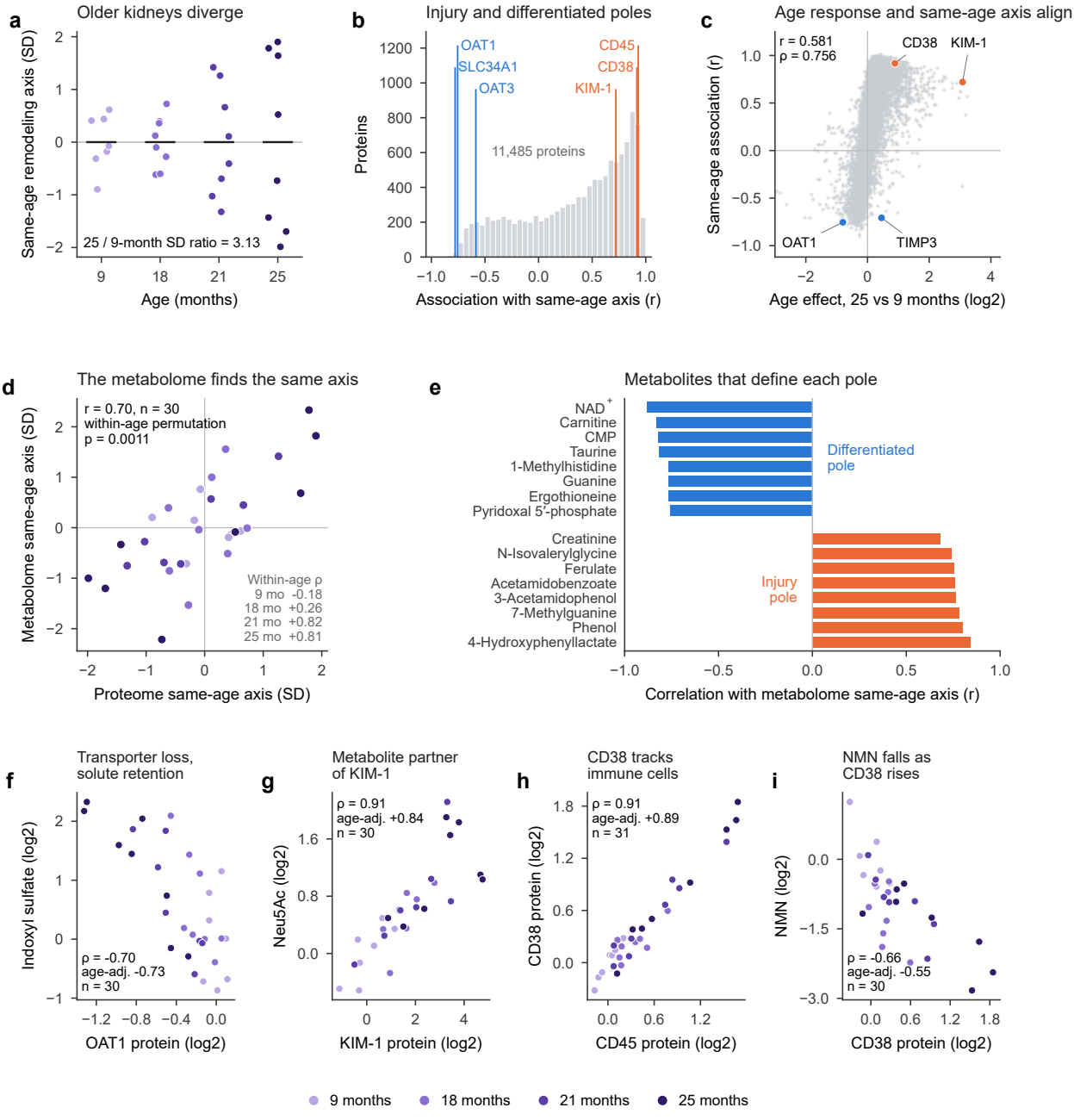

a
Older kidneys diverge
Same-age remodeling axis (SD)
25 / 9-month SD ratio = 3.13
Age (months)
b
Injury and differentiated poles
OAT1
SLC34A1
OAT3
CD45
CD38
KIM-1
11,485 proteins
Proteins
Association with same-age axis (r)
c
Age response and same-age axis align
r = 0.581
ρ = 0.756
CD38
KIM-1
OAT1
TIMP3
Same-age association (r)
Age effect, 25 vs 9 months (log2)
d
The metabolome finds the same axis
r = 0.70, n = 30
within-age permutation
p = 0.0011
Within-age ρ
9 mo -0.18
18 mo +0.26
21 mo +0.82
25 mo +0.81
Metabolome same-age axis (SD)
Proteome same-age axis (SD)
e
Metabolites that define each pole
NAD+
Carnitine
CMP
Taurine
1-Methylhistidine
Guanine
Ergothioneine
Pyridoxal 5′-phosphate
Creatinine
N-Isovalerylglycine
Ferulate
Acetamidobenzoate
3-Acetamidophenol
7-Methylguanine
Phenol
4-Hydroxyphenyllactate
Differentiated pole
Injury pole
Correlation with metabolome same-age axis (r)
f
Transporter loss, solute retention
ρ = -0.70
age-adj. -0.73
n = 30
Indoxyl sulfate (log2)
OAT1 protein (log2)
g
Metabolite partner of KIM-1
ρ = 0.91
age-adj. +0.84
n = 30
Neu5Ac (log2)
KIM-1 protein (log2)
h
CD38 tracks immune cells
ρ = 0.91
age-adj. +0.89
n = 31
CD38 protein (log2)
CD45 protein (log2)
i
NMN falls as CD38 rises
ρ = -0.66
age-adj. -0.55
n = 30
NMN (log2)
CD38 protein (log2)
9 months
18 months
21 months
25 months